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Managed tables for data lakes

Abstract

Aspects of the disclosure are directed to merging data lake openness with scalable metadata for managed tables in a cloud database platform, allowing for atomicity, consistency, isolation, and durability (ACID) transactions, performant data manipulation language (DML), higher throughput stream ingestion, data consistency, schema evolution, time travel, clustering, fine-grained security, and/or automatic storage optimization. Table data is stored in various open-source file formats in cloud storage while physical metadata of the table data is stored in a scalable metadata storage system.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application claims the benefit of the filing date of U.S. Provisional Patent Application No. 63/535,811, filed Aug. 31, 2023, the disclosure of which is hereby incorporated herein by reference.

BACKGROUND

(1) Data lakes are increasingly built using open file formats and stored on cloud object stores due to their relatively low cost and high durability. However, interfaces provided by cloud storage systems are limited to single object mutations with no support for multi-object transactions. This creates difficulty for systems using data lake storage to achieve atomicity, consistency, isolation, and durability (ACID) transactions, snapshot consistency for reads, and strong read-after-write consistency. Further, the use of open-source table formats to alleviate this difficulty causes lower write throughput, lower query performance, higher operational overhead for infrastructure management, limited transaction support, and a weaker security model.

BRIEF SUMMARY

(2) Aspects of the disclosure are directed to merging data lake openness with scalable metadata for managed tables in a cloud database platform, allowing for atomicity, consistency, isolation, and durability (ACID) transactions, performant data manipulation language (DML), higher throughput stream ingestion, data consistency, schema evolution, time travel, clustering, fine-grained security, and/or automatic storage optimization. Table data is stored in various open-source file formats in cloud storage while physical metadata of the table data is stored in a scalable metadata storage system.

(3) An aspect of the disclosure provides for a method for processing queries, including: receiving, by one or more processors, a request from a query engine to write one or more data files; writing, by the one or more processors, tuples of the one or more data files to a write-optimized storage in an appendable distributed file system; converting, by the one or more processors, the tuples to a columnar format optimized for reads; storing, by the one or more processors, the one or more data files in cloud storage compatible with the query engine; and committing, by the one or more processors, the write as an addition to a table transaction log stored in the appendable distributed file system. Another aspect of the disclosure provides for a system including: one or more processors; and one or more storage devices coupled to the one or more processors and storing instructions that, when executed by the one or more processors, cause the one or more processors to perform operations according to the method for processing queries. Yet another aspect of the disclosure provides for a non-transitory computer-readable medium for storing instructions that, when executed by one or more processors, cause the one or more processors to perform operations according to the method for processing queries.

(4) In an example, the method further includes compacting, by the one or more processors, the transaction log into columnar baselines. In another example, the columnar baselines are in a format compatible with the query engine.

(5) In yet another example, the method further includes performing, by the one or more processors, one or more maintenance tasks based on metadata in the distributed file system. In yet another example, the one or more maintenance tasks include at least one of garbage collection, file merging, file splitting, or file reclustering.

(6) In yet another example, the method further includes: converting, by the one or more processors, the table transaction log to a format compatible with the query engine; and storing, by one or more processors, the table transaction log in the cloud storage compatible with the query engine.

(7) In yet another example, the method further includes: exporting, by the one or more processors, the metadata in the appendable distributed file system to the cloud storage compatible with the query engine. In yet another example, the metadata exported is in a format compatible with the query engine. In yet another example, the exporting is automatically triggered in response to each addition to the table transaction log.

(8) In yet another example, the method further includes: receiving, by one or more processors, a request from a query engine to read the one or more data files in the cloud storage; and directly reading, by the one or more processors, the one or more data files from the cloud storage. In yet another example, the method further includes: receiving, by one or more processors, a request from a query engine to read the one or more data files in the cloud storage; and reading, by the one or more processors, the one or more data files from the cloud storage through an application programming interface (API) exposing the cloud storage and the appendable distributed file system. In yet another example, reading the one or more data files through the API guarantees at least one of exactly once delivery or strong read after write semantics.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 depicts a block diagram of an example managed table system for a cloud storage system for data lakes and data warehouses according to aspects of the disclosure.

(2) FIG. 2 depicts a block diagram of an example metadata storage system according to aspects of the disclosure.

(3) FIG. 3 depicts a block diagram of an example computing environment implementing a managed table system for a cloud storage system according to aspects of the disclosure.

(4) FIG. 4 depicts an example process for processing queries associated with writing table data according to aspects of the disclosure.

(5) FIG. 5 depicts an example process for processing queries associated with reading table data according to aspects of the disclosure.

(6) FIG. 6 depicts an example process for processing queries associated with exporting metadata according to aspects of the disclosure.

(7) FIG. 7 depicts a block diagram of an example streaming system that combines scratch space storing write-optimized data with an object store storing read-optimized data according to aspects of the disclosure.

DETAILED DESCRIPTION

(8) The technology relates generally to an approach for managed tables in a cloud database platform that can merge data lake openness with scalable metadata. The approach can allow for atomicity, consistency, isolation, and durability (ACID) transactions, performant data manipulation language (DML), higher throughput stream ingestion, data consistency, schema evolution, time travel, clustering, fine-grained security, and/or automatic storage optimization. The approach can store table data in various open-source file formats in cloud storage while storing physical metadata, e.g., file names, statistics, row-level deletion bitmaps, of the table data in a scalable metadata storage system. The separation of the physical metadata from the table data can allow for higher-volume DML and cross-table transactions.

(9) For committing a transaction, the managed table system can write file- or row-level additions and/or deletions to an append-based transaction log in the metadata storage system. The metadata storage system can include a distributed file system that supports file appends, e.g., an appendable distributed file system, to store the append-based transaction log. The distributed file system can also support snapshots, such as the ability to atomically create a logical duplicate of a group of files, and/or renames, such as the ability to atomically rename a group of files. The distributed file

system can be replicated to one or more data centers in one or more locations. The managed table system can periodically compact the append-based transaction log into read-optimized baselines to allow for improving query efficiency. The read-optimized baselines can be stored in a columnar-oriented format. For example, the managed table system can compact the append-based transaction log per minute or per hour depending on how fast the additions and/or deletions occur. The read-optimized baselines can contain column-level metadata, such as zone maps and/or bloom filters, that the managed table system can use to prune files for future queries.

(10) The managed table system can include automated data management for continuously or periodically performing maintenance tasks on the physical metadata. As examples, the maintenance tasks can include garbage collecting files, improving storage by coalescing smaller files into larger files, and/or performing continuous or periodic background reclustering.

(11) The managed table system can include a storage application programming interface (API) to support higher-throughput streaming ingestion from open-source query engines. The storage API can write ingested tuples to a write-optimized storage in the distributed file system. The write-optimized storage can be stored in a row-oriented format in an appendable file system, such as the distributed file system. The managed table system can periodically transactionally convert the ingested tuples in the write-optimized storage into read-optimized columnar data files in the cloud storage. For example, the managed table system can convert the ingested tuples per minute or per hour depending on how fast the tuples are ingested. Streamed tuples can be visible after a write as the table data can represent a union of the tuples stored in the cloud storage and the write-optimized storage in the distributed file system, allowing for maintaining atomicity guarantees of the conversion, even for cloud storage that does not support atomic renames or snapshots.

(12) The managed table system can include an open metadata view to provide a live view of the metadata storage system in open-source table formats. The open metadata view can provide compatibility with external query engines that only support open-source table formats and can avoid lock-in, as clients can export the table data and metadata. The open metadata view can be materialized to the cloud storage, such as by writing the metadata view into one or more files, or accessed through open-source compatible catalog APIs in an open-source compatible metastore. The export can be automatic or requested by running a metadata export query. The export can also be incremental, building on a previously exported snapshot and writing only the differences rather than materializing the entire metadata snapshot for each export.

(13) The open-source query engines can read table data of the managed table system by using the open metadata view to directly access files on the cloud storage. Fine-grained access control lists (ACLs) may not be enforced, as the open-source query engines have direct file access. Based on eventual consistency, recently streamed tuples still in the write-optimized storage may not be visible. Alternatively, or additionally, the open-source query engines can read the table data by reading the table data through the storage API. The storage API can enforce fine-grained ACLs and/or offer strong read consistency, regardless of whether the streamed tuple is still in the write-optimized storage.

(14) FIG. 1 depicts a block diagram of an example managed table system **100** for a cloud storage system for data lakes and data warehouses. The managed table system **100** can be implemented on one or more computing devices in one or more locations. The managed table system **100** can include one or more data lake query engines **102** and one or more data warehouse query engines **104**. Example data lake query engines **102** can include open-source engines and example data warehouse query engines **104** can include warehouse native APIs.

(15) The managed table system **100** can further include cloud storage **106** for storing table data **108** and metadata snapshots **110** in various file formats, such as open-source file formats. The metadata snapshots **110** may refer to materializing, e.g., writing into one or more files, an open metadata view of the logical metadata **114** and/or physical metadata **116** to an object store of the cloud storage **106**. The file formats can be particular to a client utilizing the managed table system **100**.

The managed table system **100** can also include a metadata storage system **112** for storing physical metadata **114** and/or logical metadata **116** in a distributed file system **118**. Example physical metadata **114** can include file names, statistics, and/or row-level deletion bitmaps of the table data **108**. Example logical metadata **116** can include schema, attributes, and/or permissions for the table data **108**. The separation of the metadata, e.g., physical metadata **114** and/or logical metadata **116**, from the table data **108** can allow for higher-volume DML and cross-table transactions.

(16) The managed table system **100** can receive requests from the data lake query engines **102** and data warehouse query engines **104** to process queries. The data warehouse query engines **104** can directly access the cloud storage **106** and metadata storage system **112** for processing the queries. The data warehouse query engines **104** can further periodically perform a disaggregated shuffle to allow workers in the data warehouse query engines **104** to exchange information, such as for distributed operations like a join operation. The data lake query engines **102** can access the cloud storage **106** and metadata storage system **112** through a storage API **122**, e.g., storage read/write API, for processing the queries. Alternatively, or additionally, the data lake query engines **102** can access the cloud storage **106** and metadata storage system **112** directly through an open metadata view (not shown).

(17) FIG. 2 depicts a block diagram of an example metadata storage system **200**. The example metadata storage system **200** can correspond to the metadata storage system **112** as depicted in FIG. 1. The metadata storage system **200** can include a distributed file system **202** for storing metadata, such as physical metadata and/or logical metadata. The distributed file system **202** can include an append-based transaction log **204** and a write-optimized storage **206** for processing transactions in response to query requests from query engines. The distributed file system **202** may further include logical metadata **208**. The query engines and a storage API can commit transactions to the metadata store system **200**. The metadata storage system **200** can further include a data management engine **210** for continuously or periodically performing maintenance on the metadata. The metadata storage system **200** can also include an open metadata view engine **212** for providing an exportable view of the metadata storage system **200** in various formats.

(18) The storage API can support high-throughput streaming ingestion based on write requests received from the data lake query engines. The storage API can write ingested tuples of a write request to the write-optimized storage **206** of the distributed file system **202**. To be write-optimized, the storage **206** can have a row-oriented format for storing tuples. Periodically, the data management engine **210** and/or the query engines can transactionally convert the ingested tuples in the write-optimized storage **206** to read-optimized data files for storing in the cloud storage. To be read-optimized, the data files can have a columnar-oriented format. As examples, based on how fast tuples are ingested, the storage API can transactionally convert the ingested tuples per minute, per hour, or continually. The tuples can be visible after a write as the table data can represent a union of data files stored in the cloud storage and data files stored in the write-optimized storage **206**. The union can allow for maintaining atomicity guarantees of the conversion, even if the cloud storage is incompatible with atomic renames or snapshots.

(19) To commit transactions based on requests received from the data lake query engines, the storage API can write file- or row-level additions and/or deletions to the append-based transaction log **204** of the distributed file system **202**. Periodically, the storage API can compact the append-based transaction log into read-optimized baselines **214**, for storing in the cloud storage. To be read-optimized, the baselines **214** can have a columnar-oriented format. As examples, based on how fast the additions and/or deletions are written, the data management engine **210** can compact the append-based transaction log **204** per minute, per hours, or continually. The read-optimized baselines **214** can contain column-level metadata that can be utilized for pruning files and/or improving query efficiency. For example, the column-level metadata can include location maps and/or bloom filters. The write-optimized storage **206**, read-optimized baselines **214**, and the append-based transaction log **204** of the distributed file system **200** allow for ACID transactions

with higher throughput.

(20) Based on requests received from the data lake query engines, the storage API can be utilized to read the table data. The storage API can enforce fine-grained access control lists (FGACs) using the logical metadata. The storage API can also offer strong consistency, regardless of whether the streamed tuple is in the cloud storage or still in the write-optimized storage since the table data can represent a union of data files of the cloud storage and write-optimized storage.

(21) The open metadata view engine **212** can provide a live view of the metadata storage system **200** in various table formats, such as open-source formats. The open metadata view engine **212** can convert the metadata to various formats and export the metadata in the various formats. For example, the open metadata view engine **212** can be configured to convert and export the metadata in a particular format compatible with a client utilizing the cloud storage. The open metadata view engine **212** can export the metadata in response to a query, such as a metadata export query received from the data lake query engines, or automatically, such as in response to each addition and/or deletion to the transaction log **204**. The open metadata view engine **212** allows for compatibility with data lake query engines or data warehouse query engines that may only support particular table formats, such as open-source table formats, as well as avoids lock-in of the metadata where the metadata may be stuck to a particular data warehouse or storage system. The open metadata view engine **212** can materialize the open metadata view to the cloud storage, such as by writing the metadata view into one or more files, or include a format-compatible API for accessing the metadata.

(22) Based on requests received from the data lake query engines, the open metadata view engine **212** can be utilized to read the table data. The open metadata view engine **212** can directly access files in the cloud storage. Since the data lake query engines may have direct file access through the open metadata view engine **212**, FGACs may not be enforced. Further, the open metadata view engine **212** can offer eventual consistency, so recently streamed tuples still in the write-optimized storage may not be visible.

(23) The data management engine **210** can perform continuous or periodic, e.g., per minute or per hour, maintenance tasks on the distributed file system, such as on the physical metadata and/or logical metadata. Maintenance tasks can include garbage collection, file coalescing, and/or reclustering, as examples. For instance, the data management engine **210** can delete files from the distributed file system that contain data that has already been deleted from the cloud storage, e.g., garbage collection of tuples stored in the write-optimized buffer **206**. As another example, the data management engine **210** can merge smaller files into larger files and/or split larger files into smaller files, such as for optimizing file sizes based on performance, compute capacity, and/or utilizing parallelism. For instance, streaming a smaller amount of data tends to produce smaller files which can then be combined and clustered to maintain performance. As yet another example, the data management engine **210** can perform background reclustering.

(24) FIG. 3 depicts a block diagram of an example computing environment **300** implementing a managed table system **302** for a cloud storage system. The managed table system **302** can correspond to the managed table system **100** as depicted in FIG. 1. The managed table system **302** can be implemented on one or more devices having one or more processors in one or more locations, such as in a server computing device **304**. A client computing device **306** and the server computing device **304** can be communicatively coupled to one or more storage devices **308** over a network **310**. The server computing device **304** and the storage devices **308** can form part of a cloud computing system **312** for cloud computing services such as Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and/or Software as a Service (SaaS).

(25) For example, the client computing device **306** may use the cloud computing system **312** as a service that provides software applications, such as accounting, word processing, inventory tracking, fraud detection, file sharing, video sharing, audio sharing, communication, or gaming. As another example, the client computing device **306** can access the cloud computing system **312** as

part of one or more operations that employ machine learning, deep learning, and/or artificial intelligence technology to train the software applications. The cloud computing system **312** can provide model parameters that can be used to update machine learning models for the software applications.

(26) The storage devices **308** can be a combination of volatile and non-volatile memory and can be at the same or different physical locations than the computing devices **304**, **306**. For example, the storage devices **308** can include any type of non-transitory computer readable medium capable of storing information, such as a hard-drive, solid state drive, tape drive, optical storage, memory card, ROM, RAM, DVD, CD-ROM, write-capable, and read-only memories.

(27) The server computing device **304** can include one or more processors **314** and memory **316**. The memory **316** can store information accessible by the processors **314**, including instructions **318** that can be executed by the processors **314**. The memory **316** can also include data **320** that can be retrieved, manipulated, or stored by the processors **314**. The memory **316** can be a type of non-transitory computer readable medium capable of storing information accessible by the processors **314**, such as volatile and non-volatile memory. The processors **314** can include one or more central processing units (CPUs), graphic processing units (GPUs), field-programmable gate arrays (FPGAs), and/or application-specific integrated circuits (ASICs), such as tensor processing units (TPUs).

(28) The instructions **318** can include one or more instructions that when executed by the processors **314**, cause the one or more processors to perform actions defined by the instructions **318**. The instructions **318** can be stored in object code format for direct processing by the processors **314**, or in other formats including interpretable scripts or collections of independent source code modules that are interpreted on demand or compiled in advance. The instructions **318** can include instructions for implementing the managed table system **302**. The managed table system **302** can be executed using the processors **314**, and/or using other processors remotely located from the server computing device **304**.

(29) The data **320** can be retrieved, stored, or modified by the processors **314** in accordance with the instructions **318**. The data **320** can be stored in computer registers, in a relational or non-relational database as a table having a plurality of different fields and records, or as JSON, YAML, proto, or XML documents. The data **320** can also be formatted in a computer-readable format such as, but not limited to, binary values, ASCII, or Unicode. Moreover, the data **320** can include information sufficient to identify relevant information, such as numbers, descriptive text, proprietary codes, pointers, references to data stored in other memories, including other network locations, or information that is used by a function to calculate relevant data.

(30) The client computing device **306** can also be configured similarly to the server computing device **304**, with one or more processors **322**, memory **324**, instructions **326**, and data **328**. The client computing device **306** can also include a client input **330** and a client output **332**. The client input **330** can include any appropriate mechanism or technique for receiving input from a client, such as keyboard, mouse, mechanical actuators, soft actuators, touchscreens, microphones, and sensors.

(31) The server computing device **304** can be configured to transmit data to the client computing device **306**, and the client computing device **306** can be configured to display at least a portion of the received data on a display implemented as part of the client output **332**. The client output **332** can also be used for displaying an interface between the client computing device **306** and the server computing device **304**. The client output **332** can alternatively or additionally include one or more speakers, transducers or other audio outputs, a haptic interface or other tactile feedback that provides non-visual and non-audible information to a client of the client computing device **306**.

(32) Although FIG. 3 illustrates the processors **314**, **322** and the memories **316**, **324** as being within the computing devices **304**, **306**, components described herein, including the processors **314**, **322** and the memories **316**, **324** can include multiple processors and memories that can operate in

different physical locations and not within the same computing device. For example, some of the instructions **318**, **326** and the data **320**, **328** can be stored on a removable SD card and other instructions within a read-only computer chip. Some or all of the instructions **31**, **326** and data **320**, **328** can be stored in a location physically remote from, yet still accessible by, the processors **314**, **322**. Similarly, the processors **314**, **322** can include a collection of processors that can perform concurrent and/or sequential operations. The computing devices **304**, **306** can each include one or more internal clocks providing timing information, which can be used for time measurement for operations and programs run by the computing devices **304**, **306**.

(33) The computing devices **304**, **306** can be capable of direct and indirect communication over the network **310**. The devices **304**, **306** can set up listening sockets that may accept an initiating connection for sending and receiving information. The network **310** itself can include various configurations and protocols including the Internet, World Wide Web, intranets, virtual private networks, wide area networks, local networks, and private networks using communication protocols proprietary to one or more companies. The network **310** can support a variety of short- and long-range connections. The short- and long-range connections may be made over different bandwidths, such as 2.402 GHz to 2.480 GHz, commonly associated with the Bluetooth® standard, 2.4 GHz and 5 GHz, commonly associated with the Wi-Fi® communication protocol; or with a variety of communication standards, such as the LTE® standard for wireless broadband communication. The network **310**, in addition or alternatively, can also support wired connections between the computing devices **304**, **306**, including over various types of Ethernet connection.

(34) Although a single server computing device **304** and user computing device **306** are shown in FIG. 3, it is understood that the aspects of the disclosure can be implemented according to a variety of different configurations and quantities of computing devices, including in paradigms for sequential or parallel processing, or over a distributed network of multiple devices. In some implementations, aspects of the disclosure can be performed on a single device, and any combination thereof.

(35) FIG. 4 depicts an example process **400** for processing queries associated with writing table data. The example process **400** can be performed on a system of one or more processors in one or more locations, such as the managed table system **100** as depicted in FIG. 1.

(36) As shown in block **410**, the managed table system **100** can be configured to receive a request from a query engine to write one or more tuples. The query engine can be a data lake query engine, such as an open-source engine.

(37) As shown in block **420**, the managed table system **100** can be configured to write the one or more tuples to a storage in a distributed file system. The storage can be a write-optimized storage having a row-oriented format for storing the tuples. The distributed file system can be an appendable distributed file system.

(38) As shown in block **430**, the managed table system **100** can be configured to convert the tuples to one or more data files in a format compatible with the query engine. The managed table system **100** can convert the tuples to a read-optimized format having a columnar-oriented format. The managed table system **100** can periodically or continually convert the tuples based on how fast the tuples are ingested. The managed table system **100** can be further configured to store the one or more data files in the cloud storage compatible with the query engine. Since table data can represent a union of data files stored in the cloud storage and storage, the tuples can be visible after a write, even if not converted yet.

(39) As shown in block **440**, the managed table system **100** can commit the write as an addition and/or deletion to a table transaction log in the distributed file system. For example, the commit can occur exactly once. The managed table system **100** can periodically or continually compact the transaction log into columnar baselines based on how fast the additions and/or deletions occur. The columnar baselines can be read-optimized and in a format compatible with the query engine. The columnar baselines can include column-level metadata.

(40) FIG. 5 depicts an example process **500** for processing queries associated with reading table data. The example process **500** can be performed on a system of one or more processors in one or more locations, such as the managed table system **100** as depicted in FIG. 1.

(41) As shown in block **510**, the managed table system **100** can be configured to receive a request from a query engine to read one or more data files in a cloud storage in a format compatible with the query engine. The query engine can be a data lake query engine, such as an open-source engine.

(42) As shown in block **520**, the managed table system **100** can be configured to determine whether to read the one or more data files directly or read the one or more data files through a storage API. The managed table system **100** can determine how to read the one or more data files based on instructions included in the request.

(43) In response to the determination, as shown in block **530**, the managed table system **100** can be configured to directly read the one or more data files. The managed table system **100** can directly access the one or more data files in the cloud storage using an open metadata view engine. Based on the direct file access, FGACs may not be enforced. Further, based on eventual consistency, recently streamed tuples in the write-optimized storage may not be visible.

(44) In response to the determination, as shown in block **540**, the managed table system **100** can be configured to read the one or more data files through the storage API. The storage API can enforce FGACs using logical metadata and offer strong consistency since the storage API can access a union of data files between the cloud storage and the write-optimized storage.

(45) FIG. 6 depicts an example process **600** for processing queries associated with exporting metadata. The example process **600** can be performed on a system of one or more processors in one or more locations, such as the managed table system **100** as depicted in FIG. 1.

(46) As shown in block **610**, the managed table system **100** can be configured to receive a request from a query engine to export metadata associated with table data in a cloud storage in a format compatible with the query engine. The query engine can be a data lake query engine, such as an open-source engine.

(47) As shown in block **620**, the managed table system **100** can be configured to convert the metadata to a format compatible with the query engine. The managed table system **100** can convert the table transaction log to a format compatible with the query engine. The managed table system **100** can convert the metadata to a particular format based on the format of the query received from the query engine.

(48) As shown in block **630**, the managed table system **100** can be configured to export the metadata. The managed table system **100** can store the table transaction log in the cloud storage compatible with the query engine.

(49) FIG. 7 depicts a block diagram of an example streaming subsystem **700**. The example streaming subsystem **700** can be included as part of the managed table system **100** as depicted in FIG. 1. The streaming subsystem **700** can be implemented on one or more computing devices in one or more locations. The streaming subsystem **700** can include a write API **702** as part of a streaming frontend, a scratch space **704**, and a streaming backend **706**. The write API **702** can correspond to the storage API **122** as depicted in FIG. 1. The scratch space **704** can be included as part of the distributed file system **118** as depicted in FIG. 1. For example, the scratch space **704** can correspond to the write-optimized buffer **206** as depicted in FIG. 2. The streaming backend **706** can correspond to the data management engine **210** as depicted in FIG. 2.

(50) The streaming subsystem **700** can provide higher throughput streaming on object stores **708** without running into the small file problem. The small file problem may refer to creating new files right after a few records are streamed, resulting in tables having a lot of tiny files, which add significant time overhead for file listing and per-file metadata overhead. These overheads limit streaming throughput and lower query performance when data is read. The object stores **708** can correspond to the cloud storage **106** as depicted in FIG. 1. The streaming subsystem **700** can utilize the scratch space **704** to bulk convert and output data in various formats, such as open source

formats, to the cloud object stores **708**.

(51) The streaming subsystem **700** can receive queries from various query engines **710**. The query engines **710** can correspond to the data lake query engines **102** and/or data warehouse query engines **104** as depicted in FIG. **1**. Based on the queries, the streaming subsystem **700** ingests data using the write API **702** and stores the data in a durable write-optimized storage, such as being stored using row-oriented format in the scratch space **704** of a distributed file system that supports appends. The write-optimized storage can be replicated to multiple locations for high availability.

(52) The streaming subsystem **700** can bulk convert data from one or more intermediate files in the scratch space **704** to a final file for storage in the object stores in various file formats. The streaming subsystem **700** can bulk convert the data using the streaming backend **706**. Metadata for the newly created object store files can be persisted in the managed table system. The query engines **710** can read data using a read API **712** immediately after a write confirmation, as the data to be read can include data from both the object stores **708** and the scratch space **704**.

(53) The streaming subsystem **700** allows for real-time insertion of row level information into object store tables via the write API **702**, such as row-by-row, in rowsets, and/or in batches. The information can be inserted, updated, and/or deleted via the write API **702** as well. The streaming subsystem **700** allows for streamed data to be immediately queried via the query engines **710** using the read API **712** after confirmation of delivery of the streamed data, even before final materialization in the object stores **708**.

(54) The managed table system can represent the union of tuples in immutable files stored in cloud object stores as well as internally stored high throughput write optimized storage. Atomicity guarantees can be offered by the conversion of files in the write optimized storage to immutable files in the cloud object stores by performing transactional commits that invalidate the write-optimized files and add read-optimized files to the managed table system. The atomicity guarantees can include exactly once delivery, where each streamed record is guaranteed to be added to the table once as well as strong read-after-write semantics, where streamed data is immediately visible to queries after write acknowledgement or commit.

(55) Aspects of this disclosure can be implemented in digital electronic circuitry, in tangibly embodied computer software or firmware, and/or in computer hardware, such as the structure disclosed herein, their structural equivalents, or combinations thereof. Aspects of this disclosure can further be implemented as one or more computer programs, such as one or more modules of computer program instructions encoded on a tangible non-transitory computer storage medium for execution by, or to control the operation of, one or more data processing apparatus. The computer storage medium can be a machine-readable storage device, a machine-readable storage substrate, a random or serial access memory device, or combinations thereof. The computer program instructions can be encoded on an artificially generated propagated signal, such as a machine-generated electrical, optical, or electromagnetic signal, which is generated to encode information for transmission to suitable receiver apparatus for execution by a data processing apparatus.

(56) The term “configured” is used herein in connection with systems and computer program components. For a system of one or more computers to be configured to perform particular operations or actions means that the system has installed thereon software, firmware, hardware, or a combination thereof that cause the system to perform the operations or actions. For one or more computer programs to be configured to perform particular operations or actions means that the one or more programs include instructions that, when executed by one or more data processing apparatus, cause the apparatus to perform the operations or actions.

(57) The term “data processing apparatus” or “data processing system” refers to data processing hardware and encompasses various apparatus, devices, and machines for processing data, including programmable processors, computers, or combinations thereof. The data processing apparatus can include special purpose logic circuitry, such as a field programmable gate array (FPGA) or an application specific integrated circuit (ASIC). The data processing apparatus can include code that

creates an execution environment for computer programs, such as code that constitutes processor firmware, a protocol stack, a database management system, an operating system, or combinations thereof.

(58) The term “computer program” refers to a program, software, a software application, an app, a module, a software module, a script, or code. The computer program can be written in any form of programming language, including compiled, interpreted, declarative, or procedural languages, or combinations thereof. The computer program can be deployed in any form, including as a standalone program or as a module, component, subroutine, or other unit suitable for use in a computing environment. The computer program can correspond to a file in a file system and can be stored in a portion of a file that holds other programs or data, such as one or more scripts stored in a markup language document, in a single file dedicated to the program in question, or in multiple coordinated files, such as files that store one or more modules, sub programs, or portions of code. The computer program can be executed on one computer or on multiple computers that are located at one site or distributed across multiple sites and interconnected by a data communication network.

(59) The term “database” refers to any collection of data. The data can be unstructured or structured in any manner. The data can be stored on one or more storage devices in one or more locations. For example, an index database can include multiple collections of data, each of which may be organized and accessed differently.

(60) The term “engine” refers to a software-based system, subsystem, or process that is programmed to perform one or more specific functions. The engine can be implemented as one or more software modules or components or can be installed on one or more computers in one or more locations. A particular engine can have one or more computers dedicated thereto, or multiple engines can be installed and running on the same computer or computers.

(61) The processes and logic flows described herein can be performed by one or more computers executing one or more computer programs to perform functions by operating on input data and generating output data. The processes and logic flows can also be performed by special purpose logic circuitry, or by a combination of special purpose logic circuitry and one or more computers.

(62) A computer or special purpose logic circuitry executing the one or more computer programs can include a central processing unit, including general or special purpose microprocessors, for performing or executing instructions and one or more memory devices for storing the instructions and data. The central processing unit can receive instructions and data from the one or more memory devices, such as read only memory, random access memory, or combinations thereof, and can perform or execute the instructions. The computer or special purpose logic circuitry can also include, or be operatively coupled to, one or more storage devices for storing data, such as magnetic, magneto optical disks, or optical disks, for receiving data from or transferring data to. The computer or special purpose logic circuitry can be embedded in another device, such as a mobile phone, a personal digital assistant (PDA), a mobile audio or video player, a game console, a Global Positioning System (GPS), or a portable storage device, e.g., a universal serial bus (USB) flash drive, as examples.

(63) Computer readable media suitable for storing the one or more computer programs can include any form of volatile or non-volatile memory, media, or memory devices. Examples include semiconductor memory devices, e.g., EPROM, EEPROM, or flash memory devices, magnetic disks, e.g., internal hard disks or removable disks, magneto optical disks, CD-ROM disks, DVD-ROM disks, or combinations thereof.

(64) Aspects of the disclosure can be implemented in a computing system that includes a back end component, e.g., as a data server, a middleware component, e.g., an application server, or a front end component, e.g., a client computer having a graphical user interface, a web browser, or an app, or any combination thereof. The components of the system can be interconnected by any form or medium of digital data communication, such as a communication network. Examples of

communication networks include a local area network (LAN) and a wide area network (WAN), e.g., the Internet.

(65) The computing system can include clients and servers. A client and server can be remote from each other and interact through a communication network. The relationship of client and server arises by virtue of the computer programs running on the respective computers and having a client-server relationship to each other. For example, a server can transmit data, e.g., an HTML page, to a client device, e.g., for purposes of displaying data to and receiving user input from a user interacting with the client device. Data generated at the client device, e.g., a result of the user interaction, can be received at the server from the client device.

(66) Unless otherwise stated, the foregoing alternative examples are not mutually exclusive, but may be implemented in various combinations to achieve unique advantages. As these and other variations and combinations of the features discussed above can be utilized without departing from the subject matter defined by the claims, the foregoing description of the embodiments should be taken by way of illustration rather than by way of limitation of the subject matter defined by the claims. In addition, the provision of the examples described herein, as well as clauses phrased as “such as,” “including” and the like, should not be interpreted as limiting the subject matter of the claims to the specific examples; rather, the examples are intended to illustrate only one of many possible embodiments. Further, the same reference numbers in different drawings can identify the same or similar elements.

Claims

1. A method for processing queries, comprising: receiving, by one or more processors, a request from a query engine to write one or more tuples; writing, by the one or more processors, the one or more tuples to a write-optimized storage in a row-oriented format in a distributed file system that supports file appends; converting, by the one or more processors, the one or more tuples to one or more data files in a columnar-oriented format compatible with the query engine; storing, by the one or more processors, the one or more data files in a read-optimized cloud storage in the columnar-oriented format compatible with the query engine; and committing, by the one or more processors, the write as an addition to a table transaction log stored in the distributed file system by writing a row-level addition to the table transaction log.
2. The method of claim 1, further comprising performing, by the one or more processors, one or more maintenance tasks in the distributed file system.
3. The method of claim 2, wherein the one or more maintenance tasks comprise garbage collection of one or more tuples stored in the write-optimized storage.
4. The method of claim 1, further comprising: converting, by one or more processors, the table transaction log to the columnar-oriented format compatible with the query engine; and storing, by one or more processors, the table transaction log in the read-optimized cloud storage compatible with the query engine.
5. The method of claim 1, wherein the query engine is a data lake query engine.
6. The method of claim 1, wherein the committing of the write to the table transaction log occurs exactly once.
7. The method of claim 1, further comprising: receiving, by the one or more processors, a request from a query engine to read the one or more data files; and reading, by the one or more processors, the one or more data files through an API that provides read after write semantics.
8. A system comprising: one or more processors; and one or more storage devices coupled to the one or more processors and storing instructions that, when executed by the one or more processors, cause the one or more processors to perform operations for processing queries, the operations comprising: receiving a request from a query engine to write one or more tuples; writing tuples of the one or more tuples to a write-optimized storage in a row-oriented format in a distributed file

system that supports file appends; converting the one or more tuples to one or more data files in a columnar-oriented format compatible with the query engine; storing the one or more data files in a read-optimized cloud storage in the columnar-oriented format compatible with the query engine; and committing the write as an addition to a table transaction log stored in the distributed file system by writing a row-level addition to the table transaction log.

9. The system of claim 8, wherein the operations further comprise performing one or more maintenance tasks in the distributed file system.

10. The system of claim 9, wherein the one or more maintenance tasks comprise garbage collection of one or more tuples stored in the write-optimized storage.

11. The system of claim 8, wherein the operations further comprise: converting the table transaction log to the columnar-oriented format compatible with the query engine; and storing the table transaction log in the read-optimized cloud storage compatible with the query engine.

12. The system of claim 8, wherein the query engine is a data lake query engine.

13. The system of claim 8, wherein the committing of the write to the table transaction log occurs exactly once.

14. The system of claim 8, wherein the operations further comprise: receiving a request from a query engine to read one or more additional data files; and reading the one or more additional data files through an API that provides read after write semantics.

15. A non-transitory computer readable medium for storing instructions that, when executed by one or more processors, cause the one or more processors to perform operations for processing queries, the operations comprising: receiving a request from a query engine to write one or more tuples; writing the one or more tuples to a write-optimized storage in a row-oriented format in a distributed file system that supports file appends; converting the one or more tuples to one or more data files in a columnar-oriented format compatible with the query engine; storing the one or more data files in a read-optimized cloud storage in the columnar-oriented format compatible with the query engine; and committing the write as an addition to a table transaction log stored in the distributed file system by writing a row-level addition to the table transaction log.

16. The non-transitory computer readable medium of claim 15, wherein the operations further comprise performing one or more maintenance tasks in the distributed file system.

17. The non-transitory computer readable medium of claim 16, wherein the one or more maintenance tasks comprise garbage collection of one or more tuples stored in the write-optimized storage.

18. The non-transitory computer readable medium of claim 15, wherein the operations further comprise: converting the table transaction log to the columnar-oriented format compatible with the query engine; and storing the table transaction log in the read-optimized cloud storage compatible with the query engine.

19. The non-transitory computer readable medium of claim 15, wherein the committing of the write to the table transaction log occurs exactly once.

20. The non-transitory computer readable medium of claim 15, wherein the operations further comprise: receiving a request from a query engine to read one or more additional data files; and reading the one or more additional data files through an API that provides read after write semantics.
